FIZZAH AMIR

Computer Science Undergraduate | Aspiring Coding Instructor
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PROFILE

Third-year Computer Science undergraduate at Information Technology University (ITU), applying to teach coding online with Code School. Curious and technical, with hands-on experience building programs in Python, C++ and JavaScript, from a Pygame arcade game and a Flutter accessibility app to a chatbot that answers student questions in real time. I enjoy breaking technical ideas into clear steps and using projects, games and examples to make coding approachable for new learners.

EDUCATION

BS Computer Science | Information Technology University (ITU), Lahore | 2024 – 2028
Data Structures and Algorithms, Advanced Database Management, Object-Oriented Programming, Network Programming

FSc, Pre-Engineering | Kinnaird College for Women, Lahore | 2022 – 2024

High School | Beaconhouse School System
Computer Science, Physics, Mathematics

SKILLS RELEVANT TO TEACHING

Programming: Python, C++, JavaScript, HTML/CSS, SQL
Beginner-friendly project tools: Pygame (game development), Flutter (mobile apps), React (basic), Git and GitHub
Core concepts: Object-Oriented Programming, Data Structures and Algorithms, Databases, REST APIs
Also: Pandas, NumPy, Scikit-learn, TensorFlow, Linux, AWS EC2, Railway Cloud

EXPERIENCE

Webmaster, Core Team Member | IEEE ITU Student Branch | Aug 2026 – Present
- Manage and maintain the society's web presence.
- Built a Python chatbot for IEEE Orientation that answers student queries in real time.

Campus Ambassador | Comeback Pakistan, ITU | Sep 2026 – Present
- Represent Comeback Pakistan's AI Accelerator Program at ITU.

PROJECTS

Space Invaders Game | May 2024 | Python, Pygame
- Recreated the classic arcade game, covering sprite management, collision detection and real-time user input.

Lumora: Color Blind App | Oct 2025 | Flutter, Postman API
- Built a mobile app that uses the device camera to detect and label colors in real time, helping color-blind users tell shades apart on everyday objects, maps and clothing.

Flight Management System | Jun 2024 – Aug 2024 | Object-Oriented Design
- Console-based system with flight scheduling, seat selection, bookings and pilot assignment, applying polymorphism, inheritance, encapsulation, aggregation and composition, with a Singleton logger and exception handling.

AeroHub Airport Management System | Dec 2025 – Jan 2026 | C++, HTML/CSS/JS, Railway Cloud
- Full-stack platform for flight scheduling, bookings, gate assignments and route optimization; B-Tree indexing and Dijkstra's algorithm for shortest routes.

Digital Wallet System API | Live on AWS EC2 | Django REST Framework, PostgreSQL
- 29 REST endpoints, 8 PostgreSQL triggers, JWT authentication, Argon2 hashing and role-based access control for 3 roles.

Insightly-PDF: RAG Document Q&A Assistant | Personal Project | Django REST Framework, React, RAG
- Full-stack app that extracts and chunks text from uploaded PDFs and answers questions with citations; Django backend for ingestion, embeddings and vector search, with a React chat frontend.

In-Memory Cache Server (Redis Clone) | ITU Course Project | C++, epoll, TCP Sockets
- Single-threaded key-value server using the Redis protocol (RESP) with crash-safe persistence; sustained 50,000+ SET/GET operations per second.

Electricity Theft Detection Using Machine Learning | ITU Group Project | Python, Scikit-learn, XGBoost, TensorFlow
- Analyzed 2M+ smart-meter records, handled class imbalance with SMOTE and benchmarked Logistic Regression, Decision Tree, Random Forest, XGBoost and a neural network.

Meeting Transcriber and Q&A Assistant (RAG) | Personal Project | Python, FAISS, Sentence Transformers
- Transcribes audio, indexes transcript embeddings in FAISS to answer questions with grounded responses, and adds summaries and action-item extraction.

CERTIFICATIONS

- Meta Backend Developer Professional Certificate, Meta (Coursera)
- Foundations of Data Science (Jun 2026), The Power of Statistics, Go Beyond the Numbers, Google (Coursera)
- Cybersecurity Bootcamp: ISO 27001:2022 (Apr 2025)

ACHIEVEMENTS

- Alibaba Hackathon: qualified for the regional round in the Open Innovation category with SOS, an emergency app.
- CodeRush, ITU (May 2025) and SOFTEC 2026: competitive programming contests.